

deliver. The Red Pro makes much more sense in a 2.5GbE or faster setup, or in larger NAS boxes where multiple users and services may be hitting the storage pool simultaneously.

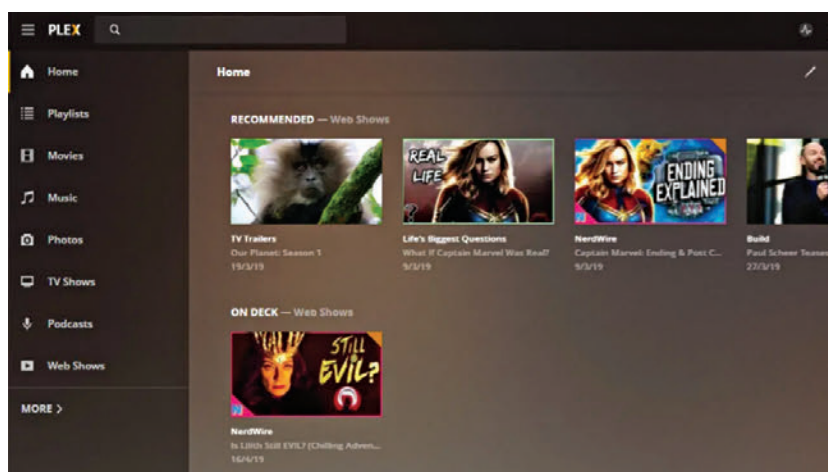
Random 4K behaviour was also good for mechanical media. It was nowhere near SSD territory, of course, but for a 7200 RPM hard drive inside a NAS, we observed some really healthy metrics. The large cache and WD's firmware behaviour appear to help with small bursty operations, particularly when the workload involves many smaller writes being absorbed before being committed to the platters.

REAL-WORLD TRANSFERS

The large-file transfer test was the cleanest demonstration of what these drives can do in a modern 2.5 GbE NAS. Copying a 50 GB 4K movie file produced an almost flat transfer graph, with speeds locked around 285 MB/s from start to finish. The transfer completed in just under three minutes, and there was no obvious drop-off once the copy operation settled.

This makes the WD Red Pro WD103KFBX a strong fit for media-heavy workflows. Large video files, RAW photo archives, project backups and system images are exactly the kind of workloads where this drive feels comfortable. The combination of high sustained sequential throughput and predictable NAS behaviour makes it feel less like a bottleneck and more like the natural storage layer behind the network connection.

Small-file performance was more variable, as expected. A 10 GB folder containing around 5,000 mixed photos and documents transferred between roughly 45 MB/s and 90 MB/s. That



is a wide range, but it is not erratic in a worrying way. Small files are simply harder for mechanical drives because the heads have to move more frequently and the file system has to process more metadata. The 512 MB cache helps smooth out some of this behaviour, but physics still applies. And the more you use the drive and data gets fragmented across the drive, the more mercurial the speeds will be.

At the end, there were no sudden freezes, no prolonged stalls and no dramatic collapse in throughput. Backing up thousands of photos, moving document archives and syncing mixed folders all felt consistent enough for a NAS of this class.

NAS APPS AND MEDIA USE

The WD Red Pro drives also behaved well when the NAS was asked to do more than simple file transfers. Plex Media Server library operations, thumbnail generation and basic timeline scrubbing were handled with minimal delay. Mechanical hard drives are not ideal for every app-heavy NAS workload, especially where databases and constant random writes are involved, but the Red Pro pair handled typical media-library behaviour without making the QTS interface feel bogged down. If you're going for a similar setup, try to look for a NAS with an SSD Caching drive.

QNAP's own background tasks, indexing and management interface

remained responsive during testing. This is where NAS-oriented firmware tuning becomes relevant. The drive has to manage command queuing and background work sensibly so that one task does not make the entire NAS feel sluggish. In this setup, the Red Pro drives handled that balance rather well. For a home media server, shared family backup system, small studio archive or office file server, this level of behaviour is exactly what you would be looking for. The drive is not pretending to be an SSD, but it does deliver enough responsiveness for a two-bay NAS that is also running everyday services.

VERDICT

The WD Red Pro WD103KFBX is a strong 10TB NAS hard drive that performs exactly the way a Pro-class NAS drive should.

—Mithun Mohandas

SPECIFICATIONS

Formatted capacity: 10TB | Recording technology: CMR | Interface: SATA | Form Factor: 3.5-inch | Drive Technology: Air | Interface transfer rate: 6 GB/s | Internal transfer rate: 237MB/s | Cache: 512 MB | RPM: 7200

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